

D428

Design Thinking for Business

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital team has interviewed patients about waiting-room stress and written a clear problem statement. The hospital's parking garage has 900 spaces. What stage comes next?

- A. Build a parking-finder app.
- B. Prototype a waiting-room redesign.
- C. Ideate: generate many possible solutions.
- D. Start over with new interviews.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

During testing, users reveal a need the team never uncovered earlier. What does design thinking suggest?

- A. Loop back to empathize or define and update the problem.
- B. Hand the issue to engineering to patch after launch.
- C. Finish the test and launch, since testing is the last stage.
- D. Discard all work and restart from nothing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A bank wants to redesign its mobile app for adults over 70. The design team, all under 30, has strong ideas. The bank's headquarters is in a historic building. Which empathize activity fits best?

- A. Have the design team brainstorm what seniors probably want.
- B. Observe and interview older adults as they use the current app at home.
- C. Tour the historic headquarters for inspiration.
- D. Survey bank executives about the app.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

After interviews, a city transit team drafts four statements. The agency's buses were repainted green last year. Which is a human-centered problem statement?

- A. Buses should be repainted a brighter color.
- B. Night-shift workers need a reliable way to know when the next bus arrives, because a missed bus means a 40-minute wait in the dark.
- C. We need to build a bus-tracking app.
- D. Increase ridership by 10% this year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team's problem statement reads: "Design a chatbot for the help desk." What is wrong with it?

- A. Nothing; naming the solution always makes a better statement.
- B. It is too focused on users.
- C. It is missing a budget figure.
- D. It names a solution instead of a user need, which limits ideas.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A grocery team learned that parents with young children feel rushed and stressed at checkout. The store has 14 checkout lanes. Which "How might we" question is best?

- A. How might we add two more lanes to reach 16?
- B. How might we make checkout feel calmer for parents with young children?
- C. How might we improve the store?
- D. How might we add self-checkout kiosks?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In an ideation session held in a room full of whiteboards, a manager critiques each idea as soon as it is shared, and people stop speaking up. Which principle is being broken?

- A. Defer judgment and aim for many ideas before narrowing.
- B. Develop only one idea at a time.
- C. Evaluate each idea immediately to save time.
- D. Switch from whiteboards to digital boards.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A team has generated 80 sticky-note ideas. What should it do next?

- A.** Build polished prototypes of all 80 ideas.
- B.** Pick the manager's favorite without criteria.
- C.** Keep generating ideas, since converging is never part of ideation.
- D.** Cluster the ideas and select the most promising ones using clear criteria.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A startup wants to test a new appointment-booking flow for a dog groomer with two locations. Its lead engineer wants to spend three months coding a full app first. Which prototype approach fits best?

- A.** The full three-month coded app.
- B.** Skip prototyping and launch directly.
- C.** A paper or clickable mock-up built in a day or two and tested with real customers.
- D.** Open a third location to test demand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

What is the main purpose of a prototype in design thinking?

- A.** To produce the final product
- B.** To impress executives with a polished demo
- C.** To prove the idea is right, every time
- D.** To learn quickly and cheaply by testing assumptions with users

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In testing, 6 of 8 users cannot find the “Book” button on a paper prototype. The team spent two weeks on custom illustrations for it. What should the team do next?

- A.** Treat it as learning; revise the layout and test again.
- B.** Conclude the users were not paying attention.
- C.** Keep the design, since the illustrations took two weeks.
- D.** Drop the idea; one failed test always means it is dead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A team’s meal-delivery drone concept is loved by users and is technically possible, but each delivery costs \$40 while customers will pay only \$8. The team’s mascot is a hummingbird. Which lens does the concept fail?

- A. None, because users love it
- B. Desirability
- C. Viability
- D. Feasibility

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A hospital team has interviewed patients about waiting-room stress and written a clear problem statement. ~~The hospital's parking garage has 900 spaces.~~ What stage comes next?

Target word: next stage **Decides:** Empathize and define are done.

A	Camouflage	Answers the garage detail, not the defined problem.
B	Dropped element	Skips ideation; no options have been explored yet.
C	Correct	Ideation follows the define stage.
D	Inverted operation	Moves backward with no new reason to.

2. Answer A Concept 1 • transfer

During testing, users reveal a need the team never uncovered earlier. What does design thinking suggest?

Target word: reverse: late finding to action **Decides:** The process is iterative.

A	Correct	New insight sends the team back to earlier stages.
B	Adjacent category	A delivery fix, not a design-thinking response to a new need.
C	Dropped element	Drops the new insight; the process is not one-way.
D	Absolute language	Iteration builds on past learning.

3. Answer B Concept 2

A bank wants to redesign its mobile app for adults over 70. The design team, all under 30, has strong ideas. ~~The bank's headquarters is in a historic building.~~ Which empathize activity fits best?

Target word: best empathy activity **Decides:** The team is not like its users.

A	Adjacent category	Relies on assumptions, not empathy.
B	Correct	Real users in real context reveal real needs.
C	Camouflage	Answers the building detail, not the users.
D	Confusion set	Executives are not the target users.

4. Answer B Concept 3

After interviews, a city transit team drafts four statements. ~~The agency's buses were repainted green last year.~~ Which is a human-centered problem statement?

Target word: best problem statement **Decides:** User, need, insight, no solution.

A	Camouflage	Answers the paint detail, not a user need.
B	Correct	Names the user, the need, and the insight.
C	Adjacent category	States a solution, not a need.
D	Confusion set	A business goal, not a user problem.

5. Answer D Concept 3 · transfer

A team's problem statement reads: "Design a chatbot for the help desk." What is wrong with it?

Target word: reverse: diagnose the statement **Decides:** It names a solution.

A	Absolute language	Locking in a solution blocks better options.
B	Inverted operation	It does not mention any user at all.
C	Adjacent category	Budget is a constraint, not the core flaw.
D	Correct	Problem statements should leave the solution open.

6. Answer B Concept 4

A grocery team learned that parents with young children feel rushed and stressed at checkout. ~~The store has 14 checkout lanes.~~ Which "How might we" question is best?

Target word: best HMW **Decides:** Balanced scope centered on the user's need.

A	Camouflage	Answers the lane count, not the parents' stress.
B	Correct	Focused on the user and need, open to many ideas.
C	Dropped element	Too broad; drops the user and the need.
D	Adjacent category	Too narrow; it already names a solution.

7. Answer A Concept 5

In an ideation session ~~held in a room full of whiteboards~~, a manager critiques each idea as soon as it is shared, and people stop speaking up. Which principle is being broken?

Target word: violated principle **Decides:** Judgment during the diverge phase.

A	Correct	Early criticism shuts down idea flow.
B	Dropped element	Drops the divergent phase entirely.
C	Inverted operation	That is what the manager is doing wrong.
D	Camouflage	The tool is not the problem.

8. Answer D Concept 5 · transfer

A team has generated 80 sticky-note ideas. What should it do next?

Target word: reverse: after diverging **Decides:** Time to converge.

A	Adjacent category	Skips selection and overspends on prototypes.
B	Dropped element	Drops the structured selection.
C	Absolute language	Ideation ends by narrowing.
D	Correct	Converging follows diverging.

9. Answer C Concept 6

A startup wants to test a new appointment-booking flow for a dog groomer ~~with two locations~~. Its lead engineer wants to spend three months coding a full app first. Which prototype approach fits best?

Target word: best prototype **Decides:** Learn fast and cheaply before building.

A	Adjacent category	High fidelity too early wastes time if assumptions are wrong.
B	Dropped element	Drops the learning step.
C	Correct	Low-fidelity prototypes speed learning.
D	Camouflage	Answers the location detail, not the booking flow.

10. Answer D Concept 6 · transfer

What is the main purpose of a prototype in design thinking?

Target word: reverse: from tool to purpose **Decides:** Prototypes exist to learn.

A	Confusion set	Prototypes are not finished products.
B	Adjacent category	Polish is not the goal.
C	Absolute language	Prototypes often show what is wrong.
D	Correct	Prototypes are learning tools.

11. Answer A Concept 7

In testing, 6 of 8 users cannot find the “Book” button on a paper prototype. ~~The team spent two weeks on custom illustrations for it.~~ What should the team do next?

Target word: response to test result **Decides:** Most users failed the core task.

A	Correct	Failure feeds the next iteration.
B	Inverted operation	Blames users instead of the design.
C	Camouflage	Sunk effort does not decide the fix.
D	Absolute language	Iteration is expected.

12. Answer C Concept 8

A team’s meal-delivery drone concept is loved by users and is technically possible, but each delivery costs \$40 while customers will pay only \$8. ~~The team’s mascot is a hummingbird.~~ Which lens does the concept fail?

Target word: failing lens **Decides:** Cost far exceeds price.

A	Dropped element	Drops the business lens.
B	Confusion set	Users love it.
C	Correct	The business cannot sustain a \$32 loss per delivery.
D	Confusion set	It can be built.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Stages and iteration	Design thinking moves through empathize, define, ideate, prototype, and test, looping back whenever new learning appears.	Treating it as a one-way checklist, or skipping a stage.	“What comes next?” or new findings late in the process.
2	Empathy from real users	Empathize by observing and talking with the actual users in their context, not by relying on the team’s assumptions.	Brainstorming what users want without meeting them.	A team that differs from the users it designs for.
3	Human-centered problem statement	A define-stage problem statement names a specific user, a need, and an insight, without naming a solution.	Writing a solution or a business target as the problem.	Statements that start with “build” or a metric.
4	“How might we” framing	A good “How might we” question is broad enough to allow many ideas but narrow enough to stay on the user’s need.	Framing too narrowly around one solution, or too broadly to guide ideas.	Several HMW questions to choose among.
5	Diverge, then converge	In ideation, first generate many ideas with judgment deferred, then cluster and select using clear criteria.	Critiquing ideas as they appear, or never narrowing down.	A brainstorm where someone judges each idea or no selection occurs.
6	Low-fidelity prototyping	Prototypes are quick, cheap artifacts built to learn and test assumptions, not finished products.	Spending months building a polished version before any user sees it.	A choice between a quick mock-up and a full build.
7	Testing as learning	Testing gathers user feedback to refine or rethink the idea; failures are information for the next iteration.	Defending the design, blaming users, or quitting at the first failure.	Users struggle with a prototype.
8	Desirability, feasibility, viability	Strong innovations sit where users want it, it can be built, and the business can sustain it.	Assuming user love alone makes an idea succeed.	An idea that passes some lenses but not others.

Course scope (public description): Design Thinking for Business examines the design thinking methodology for solving complex problems. This course introduces students to design thinking as a human-centered approach to problem-solving and innovation that draws upon empathy and creativity to develop solutions to complex problems. Students will explore the principles and stages of design thinking and analyze the use of design thinking in developing solutions through real-world scenarios.